Hypothesis Testing with E-Values

Chapter 3: Efficiency – E-values under the Alternative

Based on Ramdas & Wang

Where we are

- **Chapter 2** asked: is my e-value *valid*? ($\mathbb{E}^{\mathbb{P}}[E] \le 1$ under every null $\mathbb{P}$.)
- **Chapter 3** asks: is my e-value *any good*? I.e., under a true alternative $\mathbb{Q}$, does E actually get large?
- Validity is a *safety* property (protects against false rejections). Efficiency is a *power* property (rewards you when H_1 is true).
- Roadmap of this chapter:
    - 3.1–3.2: powered tests/e-values/p-values, and when they exist at all;
    - 3.3–3.4: the right way to measure "how good" – **e-power**;
    - 3.5–3.6: the single best possible e-variable – **log-optimality** and likelihood ratios;
    - 3.7: how to approximate the best e-variable when the alternative is composite;
    - 3.8: why $\mathbb{E}^{\mathbb{Q}}[\log E]$ is the *only* reasonable notion of e-power.

Notation you'll need I

Setup (recap from Ch. 1–2)

$\mathcal{P}$ is the set of null distributions, $\mathcal{Q}$ the set of alternative distributions. An *e-variable* $E \geq 0$ satisfies $\mathbb{E}^{\mathbb{P}}[E] \leq 1$ for every $\mathbb{P} \in \mathcal{P}$. A *p-variable* P satisfies $\mathbb{P}(P \leq t) \leq t$ for every $t \in [0,1]$, every $\mathbb{P} \in \mathcal{P}$.

Power of a test / powered e-variable (Definition 3.1)

The *power* of a test $\phi \in [0,1]$ against $\mathbb{Q}$ is $\mathbb{E}^{\mathbb{Q}}[\phi]$ (probability of correctly rejecting). A test ϕ is *powered at level* α against $\mathcal{Q}$ if

$$\mathbb{E}^{\mathbb{P}}[\phi] \leq \alpha \ \text{ for every } \mathbb{P} \in \mathcal{P}, \qquad \mathbb{E}^{\mathbb{Q}}[\phi] > \alpha \ \text{ for every } \mathbb{Q} \in \mathcal{Q}.$$

It is *uniformly* powered if $\inf_{\mathbb{Q} \in \mathcal{Q}} \mathbb{E}^{\mathbb{Q}}[\phi] > \alpha$ (a single margin works for the whole alternative set).
Likewise E is a *powered e-variable* against $\mathcal{Q}$ if $\mathbb{E}^{\mathbb{Q}}[E] > 1$ for every $\mathbb{Q} \in \mathcal{Q}$.
Analogy: think of ϕ as a smoke alarm. "Powered" means it rarely false-alarms (controlled by α) *and* it reliably goes off when there really is a fire (every $\mathbb{Q}$).

Notation you'll need II

E-power (Definition 3.11) – today's central new quantity

The *e-power* of an e-variable E against $\mathbb{Q}$ is

$$\mathbb{E}^{\mathbb{Q}}[\log E].$$

Plain English: it is the *expected growth rate, on a log scale*, of your evidence when the alternative $\mathbb{Q}$ is really true. Since e-values are naturally *multiplied* across independent experiments, the log turns multiplication into addition – exactly like compound interest, where you track $\log(\text{wealth})$ to get a steady per-period growth *rate*. Higher e-power $=$ your evidence compounds faster $=$ a better e-variable.

Powered e-value/test, restated with e-power in mind

E has *positive e-power* against $\mathcal{Q}$ if $\mathbb{E}^{\mathbb{Q}}[\log E] > 0$ for every $\mathbb{Q} \in \mathcal{Q}$; *uniformly* positive if $\inf_{\mathbb{Q} \in \mathcal{Q}} \mathbb{E}^{\mathbb{Q}}[\log E] > 0$. Positive e-power is a *stronger* requirement than being a powered e-variable ($\mathbb{E}^{\mathbb{Q}}[E] > 1$), by Jensen's inequality (Section 3.3 below).

Notation you'll need III

Log-optimality (Definition 3.18)

Comparing two e-variables E_1, E_2 for the same $\mathbb{P}$ vs. simple $\mathbb{Q}$: E_1 has *greater e-power* than E_2 if $\mathbb{E}^{\mathbb{Q}}[\log(E_2/E_1)] \leq 0$. An e-variable E is *log-optimal* if $\mathbb{E}^{\mathbb{Q}}[\log(E'/E)] \leq 0$ for every other e-variable E'. *Analogy:* among all legal betting strategies against a rigged coin, the log-optimal one is the *Kelly bet* – the one whose long-run bankroll growth rate cannot be beaten by any other strategy. Theorem 3.20 below shows this best strategy is simply the likelihood ratio $\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P}$.

Notation you'll need IV

Kullback–Leibler (KL) divergence – defined plainly

For distributions $\mathbb{Q}, \mathbb{P}$ with densities q, p w.r.t. a common reference measure $\mathbb{L}$,

$$\mathrm{KL}(\mathbb{Q} \,\|\, \mathbb{P}) \;:=\; \mathbb{E}^{\mathbb{Q}}\left[\log \frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\right] \;=\; \int q(x) \log \frac{q(x)}{p(x)}\, \mathbb{L}(\mathrm{d}x).$$

Plain-English meaning: $\mathrm{KL}(\mathbb{Q} \,\|\, \mathbb{P})$ is the *average log-likelihood-ratio* of $\mathbb{Q}$ over $\mathbb{P}$, when data really come from $\mathbb{Q}$ – a measure of *how distinguishable* $\mathbb{Q}$ *is from* $\mathbb{P}$. It is always ≥ 0 (Gibbs' inequality), and $= 0$ exactly when $\mathbb{P} = \mathbb{Q}$ (indistinguishable). Bigger KL $=$ easier to tell $\mathbb{Q}$ apart from $\mathbb{P}$ $=$ faster evidence accumulation.

Why this matters together

Theorem 3.20 (Section 3.5) shows: the log-optimal e-variable for simple $\mathbb{P}$ vs. simple $\mathbb{Q}$ is exactly the likelihood ratio $\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P}$, and its (best possible) e-power equals $\mathrm{KL}(\mathbb{Q} \,\|\, \mathbb{P})$. This single fact will anchor our running numerical example.

Section 3.1 – Intuition: what does "powered" mean?

- A test/e-variable is *valid* if it rarely cries wolf under H_0.
- It is *powered* if, in addition, it reliably flags H_1 when H_1 is true.
- Crucially: powered-ness is a property of a *pair* $(\mathcal{P}, \mathcal{Q})$. If the alternative $\mathcal{Q}$ creeps arbitrarily close to the null $\mathcal{P}$ (a "boundary point"), no test or e-variable can be powered – there is no way to reliably distinguish two hypotheses that are nearly identical.

Definition 3.1: powered tests and e-values

The *power* of a test ϕ against $\mathbb{Q}$ is $\mathbb{E}^{\mathbb{Q}}[\phi]$; the *type-II error rate* is $1-$ power. Testing null $\mathcal{P}$, a test ϕ is *powered at level* $\alpha \in (0,1)$ against $\mathcal{Q}$ if

$$\mathbb{E}^{\mathbb{P}}[\phi] \leq \alpha \;\; \forall \mathbb{P} \in \mathcal{P}, \qquad \mathbb{E}^{\mathbb{Q}}[\phi] > \alpha \;\; \forall \mathbb{Q} \in \mathcal{Q},$$

and *uniformly* powered if further $\inf_{\mathbb{Q}\in\mathcal{Q}} \mathbb{E}^{\mathbb{Q}}[\phi] > \alpha$. An e-variable E is *powered* against $\mathcal{Q}$ if $\mathbb{E}^{\mathbb{Q}}[E] > 1$ for every $\mathbb{Q} \in \mathcal{Q}$, and *uniformly* powered if $\inf_{\mathbb{Q}\in\mathcal{Q}} \mathbb{E}^{\mathbb{Q}}[E] > 1$.

3.1 Powered tests, e-values, p-values II

Example 3.3: when powered tests don't exist

Let $X_1, \dots, X_n$ be iid $N(\mu, 1)$. Testing $H_0 : \mu \leq 0$ vs. $H_1 : \mu > 0$: the e-variable $\exp(X_1 + \cdots + X_n - n/2)$ *is* powered (though not uniformly). But testing $H_0 : \mu < 0$ vs. $H_1 : \mu \geq 0$: **no** powered test or e-variable exists at all! The reason: $\mu = 0$ is a boundary point shared in spirit by both sides, so nothing can have power there while staying valid arbitrarily close to it (proof uses Fatou's lemma to push a valid procedure at $\mu = -1/m$ through to the limit $\mu = 0$).
Takeaway (Remark 3.4): if H_0 is an open set and H_1 is its exact complement, with continuous densities, powered procedures generically do not exist. Always leave an explicit gap around the boundary in practice.

Definition 3.5: all-or-nothing e-variable

E is *all-or-nothing* if $E = \mathbf{1}_{\{A\}}/c$ for some event A and constant $c > 0$: it takes only the values 0 or $1/c$. These are exactly the e-variables that come from "hard" binary tests.

3.1 Powered tests, e-values, p-values III

Fact 3.6: tests $\leftrightarrow$ bounded e-variables

For any $\alpha \in (0,1)$, there is a one-to-one correspondence between $[0, 1/\alpha]$-valued e-variables and level-α tests, via

$$E = \phi/\alpha.$$

If ϕ is binary (reject iff a p-value $P \le \alpha$), the corresponding $E = \mathbf{1}_{\{P \le \alpha\}}/\alpha$ is all-or-nothing.
Consequence: a powered level-α binary test exists iff a powered e-variable taking values in $\{0, 1/\alpha\}$ exists – e-values can always replicate what p-value tests do.

Definition 3.7: powered p-values

For $\beta \in (0,1]$, a p-variable P is *powered on* $[0,\beta]$ against $\mathcal{Q}$ if $\mathbb{Q}(P \le t) \ge t$ for all $t \in [0,\beta]$ (with strict inequality somewhere). It is *powered* if this holds with $\beta = 1$, meaning P is strictly stochastically smaller than Uniform$[0,1]$ under every $\mathbb{Q} \in \mathcal{Q}$. In practice $\beta = 0.25$ already suffices, since real rejections happen at small α.

Python: powered vs. not-powered (Example 3.3)

```python
import numpy as np

def lr_evalue(x, mu):
    """Likelihood-ratio e-variable for H0: N(0,1) vs shift mu, n obs."""
    n = len(x)
    return np.exp(mu * x.sum() - n * mu**2 / 2)

rng = np.random.default_rng(0)
n = 50
# H1: mu > 0 strictly -- a powered e-variable exists, e.g. mu_hat = 1
mu_true = 1.0
reps = 20000
x = rng.normal(mu_true, 1, size=(reps, n))
E = np.array([lr_evalue(xi, 1.0) for xi in x])
print("mean E under Q (mu=1):", E.mean())          # > 1, as guaranteed
print("fraction E > 1/0.05 :", (E > 20).mean())  # power at level 0.05

# H0: mu < 0  vs  H1: mu >= 0 -- NO powered e-variable can exist
# (boundary point mu=0 is shared "in the limit" -- see Example 3.3 proof)
```

Running this confirms $\mathbb{E}^{\mathbb{Q}}[E] > 1$ numerically – but for the boundary-touching version of the problem, no choice of E would ever give a uniform margin, however many observations we simulate.

3.2 A unified existence result I

Assumption (U)

There exists $U \stackrel{d}{=} \mathrm{Unif}[0,1]$ under every $\mathbb{P} \in \mathcal{P}$ and $\mathbb{Q} \in \mathcal{Q}$, independent of everything else. (Very weak – always satisfiable by "lifting" the sample space with an auxiliary coin flip.)

Theorem 3.9: five equivalent notions of powered-ness

Under (U), for testing $\mathcal{P}$ against $\mathcal{Q}$ (no restriction), the following are equivalent:

(i) a bounded powered e-variable exists;

(ii) a powered level-α test exists (for some/every $\alpha \in (0,1)$);

(iii) a powered level-α *binary* test exists (for some/every α);

(iv) a p-variable powered on $[0,\beta]$ exists (for some/every $\beta \in (0,1)$);

(v) a powered p-variable exists.

3.2 A unified existence result II

Intuition: this is a “no free lunch, no hidden extra power” result. It says testability is testability, however you package it – as a test, a bounded e-variable, or a p-variable. You cannot get more discriminating power by switching representations. (The directions (i)⇔(ii), (iii)⇒(i), (v)⇒(i) even hold without (U).)

Example 3.10: finite sample spaces are subtle

On a finite Ω with $\mathbb{P}(\{\omega\}) > 0$ everywhere, testing $\mathbb{P}$ vs. $\mathbb{Q} \neq \mathbb{P}$: a powered e-variable is simply the likelihood ratio $\mathbb{Q}/\mathbb{P}$. But *no* p-variable can be powered on $[0, \beta]$ for any β – because any p-variable on a finite space takes only finitely many values, so $\mathbb{Q}(P \leq \alpha) = 0$ for small enough $\alpha > 0$, violating Definition 3.7. **Lesson:** e-variables can succeed where powered p-variables structurally cannot.

Section 3.3 – Intuition: why e-power, not just $\mathbb{E}^{\mathbb{Q}}[E] > 1$?

- "Powered" ($\mathbb{E}^{\mathbb{Q}}[E] > 1$) only says the e-variable is valuable *on average*, in raw (arithmetic) terms.
- But e-values are designed to be *multiplied* across repeated/sequential experiments: $M_t = \prod_{s=1}^{t} E_s$.
- What matters for M_t's long-run behavior is the average of $\log E_s$, not of E_s – exactly as in compound-interest/Kelly-betting problems.
- This motivates **e-power** $= \mathbb{E}^{\mathbb{Q}}[\log E]$: the exponential growth rate of your cumulative evidence.

Definition 3.11: e-power

The *e-power* of E against $\mathbb{Q}$ is $\mathbb{E}^{\mathbb{Q}}[\log E]$ (assumed well-defined; it may equal $+\infty$ or $-\infty$). E has *positive e-power* against $\mathcal{Q}$ if $\mathbb{E}^{\mathbb{Q}}[\log E] > 0$ for each $\mathbb{Q} \in \mathcal{Q}$; *uniformly* positive if $\inf_{\mathbb{Q}\in\mathcal{Q}} \mathbb{E}^{\mathbb{Q}}[\log E] > 0$.

Why the log? Connection to growth rate (Kelly criterion)

For iid e-variables $E_1, E_2, \ldots$ with finite e-power, let $M_t = \prod_{s=1}^{t} E_s$, $t \in \mathbb{N}$. By the strong law of large numbers,

$$\frac{\log M_t}{t} \xrightarrow{\mathbb{Q}\text{-a.s.}} \mathbb{E}^{\mathbb{Q}}[\log E_1] \qquad (t \to \infty),$$

i.e. the e-power *is* the asymptotic growth rate of the cumulative e-process. This is exactly the classical Kelly criterion from betting/investment theory.

Proposition 3.12: e-power controls power

Let $E_1, E_2, \dots$ be iid under $\mathbb{Q}$, $M_t = \prod_{s=1}^t E_s$. If $\mathbb{E}^{\mathbb{Q}}[\log E_1] > 0$, then $\mathbb{Q}(M_t > 1/\alpha) \to 1$ as $t \to \infty$, for every $\alpha > 0$ (asymptotic power 1). If $\mathbb{E}^{\mathbb{Q}}[\log E_1] < 0$, then $\mathbb{Q}(M_t > 1/\alpha) \to 0$ (asymptotic power 0).

3.3 E-power and powered e-values III

Example 3.13: Gaussian e-power grows linearly in n

Testing $\mathbb{P} = N(0,1)$ vs. $\mathbb{Q} = N(\mu,1)$, n iid observations. The likelihood-ratio e-value is

$$E_n = \prod_{i=1}^{n} \exp(\mu X_i - \mu^2/2),$$

with e-power $\mathbb{E}^{\mathbb{Q}}[\log E_n] = n\,\mathbb{E}^{\mathbb{Q}}[\mu X_1 - \mu^2/2] = n\mu^2/2$: positive, and *linear in* n – more data means faster-growing evidence.

3.3 E-power and powered e-values IV

Theorem 3.14: powered vs. positive e-power (they're closely related)

For nonnegative X and any $\mathbb{Q}$,

$$\mathbb{E}^{\mathbb{Q}}[X] > 1 \iff \mathbb{E}^{\mathbb{Q}}[\log(1-\lambda+\lambda X)] > 0 \text{ for some } \lambda \in [0,1].$$

So "powered" ($\mathbb{E}^{\mathbb{Q}}[X] > 1$) is *equivalent* to *some soft-clipped version* of X having positive e-power. Geometrically: $\mathbb{E}^{\mathbb{Q}}[\log E] > 0$ means the *geometric* mean of E exceeds 1, a strictly stronger statement than the *arithmetic* mean $\mathbb{E}^{\mathbb{Q}}[E] > 1$ (Jensen's inequality). From any powered e-variable E, the shrunk version $1-\lambda+\lambda E$ (small $\lambda > 0$) has positive e-power.

Example 3.15: pathological gap between the two notions

Let $\Omega = \mathbb{N}_0$, $\mathbb{P} = \frac{1}{2}\delta_0 + \frac{1}{2}\delta_1$, $\mathcal{Q} = \{\mathbb{Q}_n\}$ with $\mathbb{Q}_n = \frac{1}{n}\delta_n + (1-\frac{1}{n})\delta_0$. Taking $E(n) = 2n$ gives a *uniformly powered, unbounded* e-variable ($\mathbb{E}^{\mathbb{Q}}_n[E] = 2$ for all n), yet $\inf_n \mathbb{E}^{\mathbb{Q}_n}[\log(1-\lambda+\lambda E)] < 0$ for every λ – and in fact *no bounded* e-variable can even be powered here. Uniform power alone does not guarantee good (or even boundable) e-power.

Example 3.16: e-power can mislead in the tails

Let Y be Pareto(1) under $\mathbb{Q}$ (infinite mean) and $E = \exp(6 - Y/100)$. Then $\mathbb{E}^{\mathbb{Q}}[\log E] = -\infty$, yet $\mathbb{Q}(E > 100) > 0.99$: E gives very strong evidence with high probability, despite having "the worst possible" e-power. Moral: e-power is the right *long-run, repeated-trial* criterion (Prop. 3.12), but is not the only lens for judging a single-shot e-value.

Python: e-power vs. raw power, simulated (Example 3.13)

```python
import numpy as np
rng = np.random.default_rng(1)

def simulate_epower(mu, n, reps=200000):
    X = rng.normal(mu, 1, size=(reps, n))
    logE = mu * X.sum(axis=1) - n * mu**2 / 2    # log of LR e-value
    return logE.mean(), np.exp(logE).mean()      # (e-power, plain E[E])

for n in [1, 10, 50, 100]:
    epow, meanE = simulate_epower(mu=0.3, n=n)
    print(f"n={n:4d}  e-power~{epow:7.3f}  (theory {n*0.3**2/2:7.3f})  "
          f"E[E]~{meanE:8.2f}")
```

Output confirms e-power $\approx n\mu^2/2$ growing *linearly*, while the raw mean $\mathbb{E}^{\mathbb{Q}}[E]$ grows *exponentially* in n – both say "more data is better," but e-power is the additive, per-observation quantity that matches the Kelly growth-rate interpretation.

3.4 Conditions for existence of powered e-variables I

Total variation (TV) distance

For distributions $\mathbb{P}, \mathbb{Q}$: $d_{\mathrm{TV}}(\mathbb{P}, \mathbb{Q}) = \sup_A |\mathbb{P}(A) - \mathbb{Q}(A)|$ (biggest possible probability gap on any event). For sets of distributions, $d_{\mathrm{TV}}(\mathcal{P}, \mathcal{Q}) = \inf_{\mathbb{P}\in\mathcal{P}, \mathbb{Q}\in\mathcal{Q}} d_{\mathrm{TV}}(\mathbb{P}, \mathbb{Q})$: think of it as "how close the nearest null scenario can get to the nearest alternative scenario."

Kraft–Le Cam theorem (background)

With a common reference measure: there is a test ϕ with a strict power/size gap $\inf_{\mathbb{Q}} \mathbb{E}^{\mathbb{Q}}[\phi] - \sup_{\mathbb{P}} \mathbb{E}^{\mathbb{P}}[\phi] > 2\varepsilon$ if and only if $d_{\mathrm{TV}}(\mathrm{Conv}(\mathcal{P}), \mathrm{Conv}(\mathcal{Q})) > 2\varepsilon$ (convex hulls, in TV distance).

3.4 Conditions for existence of powered e-variables II

Theorem 3.17: a checkable existence criterion

For testing $\mathcal{P}$ against $\mathcal{Q}$ with a common reference measure, TFAE:

(i) some test ϕ has $\inf_{\mathbb{Q}\in\mathcal{Q}} \mathbb{E}^{\mathbb{Q}}[\phi] > \sup_{\mathbb{P}\in\mathcal{P}} \mathbb{E}^{\mathbb{P}}[\phi]$;

(ii) a uniformly powered *bounded* e-variable exists for $\mathcal{P}$ against $\mathcal{Q}$;

(iii) $d_{\mathrm{TV}}(\mathrm{Conv}(\mathcal{P}), \mathrm{Conv}(\mathcal{Q})) > 0$.

Intuition: you can be uniformly powered (with a bounded e-variable, no less) *if and only if* the convex hulls of your null and alternative scenario-sets are genuinely, positively separated in total variation. This turns an abstract testability question into a concrete geometric one about two convex sets of distributions. When $\mathcal{P}, \mathcal{Q}$ are finite, (iii) reduces to the two convex hulls simply being disjoint: $\mathrm{Conv}(\mathbb{Q}_1, \ldots, \mathbb{Q}_M) \cap \mathrm{Conv}(\mathbb{P}_1, \ldots, \mathbb{P}_L) = \varnothing$.

Running example, part 1: testing a coin's fairness

Setup

$H_0 : \theta = \theta_0 = 0.5$ (fair coin) vs. $H_1 : \theta = \theta_1 = 0.7$ (biased coin). Observe n iid flips $X_1, \ldots, X_n \in \{0, 1\}$ ($X_i = 1$ for heads).

The likelihood-ratio e-variable, one flip at a time

$$E_i = \frac{\theta_1^{X_i}(1-\theta_1)^{1-X_i}}{\theta_0^{X_i}(1-\theta_0)^{1-X_i}} = \left(\tfrac{0.7}{0.5}\right)^{X_i}\left(\tfrac{0.3}{0.5}\right)^{1-X_i} = 1.4^{X_i} \cdot 0.6^{1-X_i}.$$

So $E_i = 1.4$ on heads, $E_i = 0.6$ on tails – and $E_n = \prod_{i=1}^n E_i$ over n flips is a valid e-variable for H_0 (check: $\mathbb{E}^{\theta_0}[E_i] = 0.5(1.4) + 0.5(0.6) = 1$, exactly).

Step 1: e-power under the true alternative $\theta_1 = 0.7$

$$\mathbb{E}^{\theta_1}[\log E_i] = 0.7\log(1.4) + 0.3\log(0.6) = 0.7(0.33647) + 0.3(-0.51083) = 0.08228.$$

This equals $\mathrm{KL}(\mathrm{Bern}(0.7) \,\|\, \mathrm{Bern}(0.5)) = 0.08228$ nats – no coincidence (Sec. 3.5): the LR e-variable is log-optimal, with e-power always equal to the KL divergence.

Section 3.5 – Intuition: what's the *best possible* e-variable?

- For a simple null $\mathbb{P}$ vs. simple alternative $\mathbb{Q}$, classical (Neyman–Pearson) theory says the best *test* thresholds the likelihood ratio $q(x)/p(x)$.
- Question: is there an analogous "best e-variable"? Answer: yes, and it is even simpler – it is *exactly* the likelihood ratio itself, with no thresholding needed.
- Why? An e-variable just needs $\mathbb{E}^{\mathbb{P}}[E] \leq 1$; the likelihood ratio $\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P}$ satisfies this with equality, and no other e-variable can have larger e-power.

3.5 Log-optimality and likelihood ratios I

Definition 3.18: log-optimality (restated)

E_1 has greater (resp. equal) e-power than E_2 if $\mathbb{E}^{\mathbb{Q}}[\log(E_2/E_1)] \le 0$ (resp. $= 0$). E is *log-optimal* if $\mathbb{E}^{\mathbb{Q}}[\log(E'/E)] \le 0$ for every other e-variable E' – i.e., no other e-variable can beat its e-power.

Proposition 3.19: existence and uniqueness

For any $\mathcal{P}$, a log-optimal e-variable against $\mathbb{Q}$ always exists and is $\mathbb{Q}$-almost-surely unique.

Kullback–Leibler divergence, Eq. (3.4)

$$\mathrm{KL}(\mathbb{Q} \,\|\, \mathbb{P}) := \mathbb{E}^{\mathbb{Q}}\left[\log \frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\right] = \int q(x) \log \frac{q(x)}{p(x)} \, \mathbb{L}(\mathrm{d}x) \quad (= \infty \text{ if } \mathbb{Q} \not\ll \mathbb{P}).$$

Gibbs' inequality: $\mathrm{KL}(\mathbb{Q} \,\|\, \mathbb{P}) \ge 0$ always, with equality iff $\mathbb{P} = \mathbb{Q}$.

3.5 Log-optimality and likelihood ratios II

Theorem 3.20: THE key result of this section

For testing simple $\mathbb{P}$ against simple $\mathbb{Q}$ with $\mathbb{Q} \ll \mathbb{P}$: the ($\mathbb{P}$-a.s. unique) log-optimal e-variable is the likelihood ratio $\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P}$. Its e-power equals $\mathrm{KL}(\mathbb{Q} \,\|\, \mathbb{P})$ (possibly ∞).

3.5 Log-optimality and likelihood ratios III

Step-by-step: why the likelihood ratio wins

Step 1. Any exact e-variable can be written $E = \mathrm{d}\mathbb{S}/\mathrm{d}\mathbb{P}$ for some distribution $\mathbb{S} \ll \mathbb{P}$ (Proposition 2.12), and any e-variable is dominated by an exact one, so it suffices to compare exact e-variables.
Step 2. It further suffices to take $\mathbb{S} \ll \mathbb{Q}$ (else replace $\mathbb{S}$ by its $\mathbb{Q}$-absolutely-continuous part; e-power is unaffected $\mathbb{Q}$-a.s.).
Step 3. Compare $E = \mathrm{d}\mathbb{S}/\mathrm{d}\mathbb{P}$ against the likelihood ratio $\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P}$:

$$\mathbb{E}^{\mathbb{Q}}\left[\log \frac{E}{\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P}}\right] = \int q(x) \log\left(\frac{s(x)/p(x)}{q(x)/p(x)}\right) \mathbb{L}(\mathrm{d}x) = -\int q(x) \log \frac{q(x)}{s(x)} \mathbb{L}(\mathrm{d}x) = -\mathrm{KL}(\mathbb{Q} \,\|\, \mathbb{S}) \leq 0,$$

using Gibbs' inequality $\mathrm{KL}(\mathbb{Q} \,\|\, \mathbb{S}) \geq 0$. **Conclusion:** $\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P}$ beats every candidate E, so it is log-optimal; equality holds only when $\mathbb{S} = \mathbb{Q}$, giving uniqueness.

Example 3.21: confirming optimality in the Gaussian case

Continuing Example 3.13: for $E_n(\theta) = \prod_i \exp(\theta X_i - \theta^2/2)$ (any fixed guess θ, not necessarily μ), the e-power under true $\mathbb{Q} = N(\mu, 1)^n$ is

$$\mathbb{E}^{\mathbb{Q}}[\log E_n(\theta)] = n(\theta\mu - \theta^2/2).$$

Maximizing over θ: derivative $\mu - \theta = 0 \Rightarrow \theta^\star = \mu$, confirming Theorem 3.20 – guessing the *true* mean shift is exactly optimal, with maximal e-power $n\mu^2/2 = n \cdot \mathrm{KL}(N(\mu,1) \,\|\, N(0,1))$.

Running example, part 2: log-optimality in numbers

Step 1: the log-optimal e-variable IS the likelihood ratio

By Theorem 3.20, $E_i = 1.4^{X_i} 0.6^{1-X_i}$ (the LR e-variable from before) is log-optimal for $H_0 : \theta = 0.5$ vs. $H_1 : \theta = 0.7$, and its best-possible e-power equals

$$\mathrm{KL}(\mathrm{Bern}(0.7) \,\|\, \mathrm{Bern}(0.5)) = 0.7 \log \tfrac{0.7}{0.5} + 0.3 \log \tfrac{0.3}{0.5} = 0.08228 \text{ nats.}$$

Step 2: a naive (suboptimal) e-variable

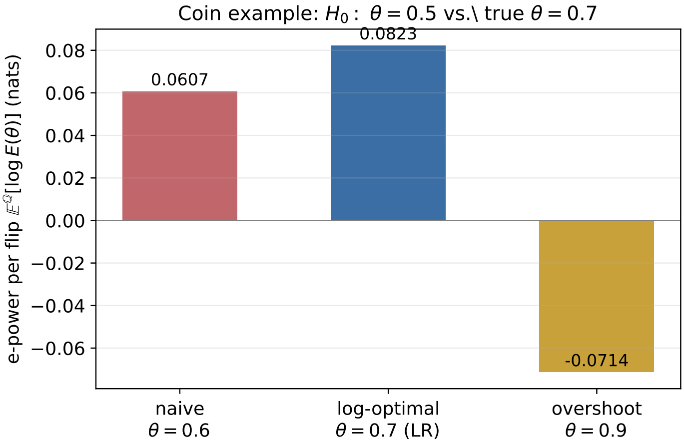
Suppose instead we guess a too-conservative alternative $\theta = 0.6$: $E_i' = 1.2^{X_i} \cdot 0.8^{1-X_i}$. Its e-power under the true $\theta_1 = 0.7$ is

$$\mathbb{E}^{0.7}[\log E_i'] = 0.7 \log(1.2) + 0.3 \log(0.8) = 0.7(0.18232) + 0.3(-0.22314) = 0.06068 \text{ nats.}$$

Step 3: compare growth over $n = 100$ flips

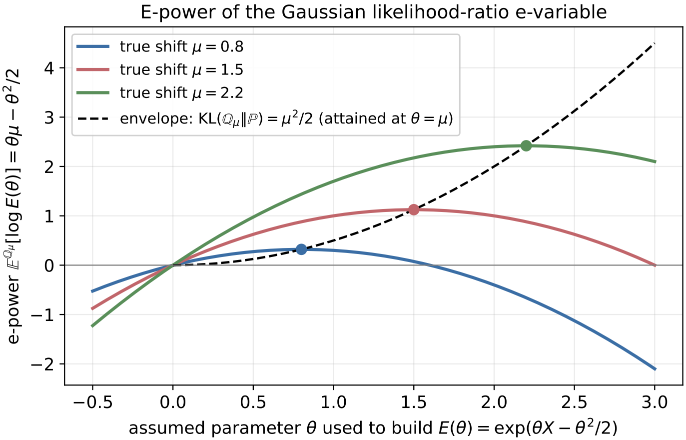
Log-optimal: e-power $\times 100 = 8.228$, so $M_{100} \approx e^{8.228} \approx 3{,}740$. Naive: e-power $\times 100 = 6.068$, so $M_{100} \approx e^{6.068} \approx 432$. *The log-optimal e-variable compounds nearly* $9\times$ *faster over 100 flips.*

Figure: e-power in the coin example, three choices of θ



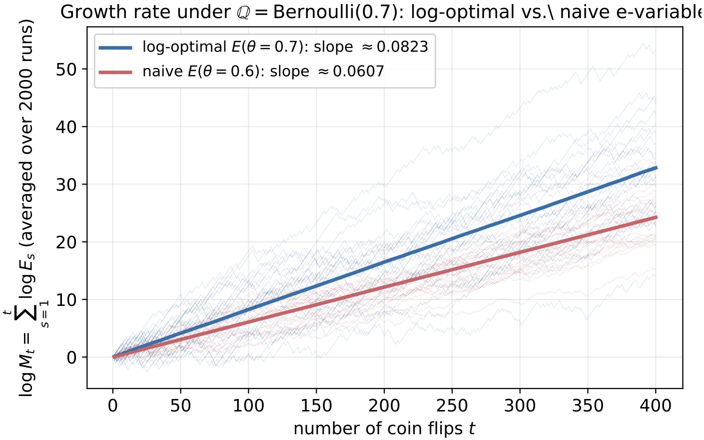
Per-flip e-power $\mathbb{E}^{\theta_1=0.7}[\log E(\theta)]$ for three assumed parameters θ: the naive under-guess ($\theta = 0.6$), the log-optimal true value ($\theta = 0.7$), and an overshoot ($\theta = 0.9$). The log-optimal choice always wins (Theorem 3.20).

Figure: e-power as a function of the assumed parameter



For the Gaussian LR e-variable $E(\theta) = \exp(\theta X - \theta^2/2)$ under true shift μ: e-power $= \theta\mu - \theta^2/2$ peaks exactly at $\theta = \mu$ (dots), touching the KL envelope $\mu^2/2$ (dashed) – a visual proof of Theorem 3.20 / Example 3.21.

Figure: growth rate over repeated coin flips



$\log M_t = \sum_{s<t} \log E_s$ under the true $\mathbb{Q} = \mathrm{Bernoulli}(0.7)$: log-optimal $(\theta = 0.7)$ climbs with slope 0.0823, the naive choice $(\theta = 0.6)$ with slope 0.0607 – matching Proposition 3.12's growth-rate interpretation of e-power.

Python: log-optimality & KL divergence, from scratch

```python
import numpy as np

theta0, theta1_true, theta_naive = 0.5, 0.7, 0.6

def kl_bernoulli(q, p):
    """KL(Bernoulli(q) || Bernoulli(p)), in nats."""
    return q*np.log(q/p) + (1-q)*np.log((1-q)/(1-p))

epower_optimal = kl_bernoulli(theta1_true, theta0)          # = KL(0.7||0.5)
epower_naive   = kl_bernoulli(theta1_true, theta_naive)     # wrong-target LR

print(f"log-optimal e-power per flip : {epower_optimal:.5f} nats")
print(f"naive e-power per flip       : {epower_naive:.5f} nats")
print(f"ratio of growth after n=100  : "
      f"{np.exp(100*epower_optimal)/np.exp(100*epower_naive):.2f}x")
# -> log-optimal e-power per flip : 0.08228 nats
#    naive e-power per flip       : 0.06068 nats
#    ratio of growth after n=100  : 8.66x
```

3.6 Conditional log-optimality & log-optimal calibrators I

- In practice, an e-variable must be computable from the *observed data*, i.e. measurable w.r.t. some sub-σ-algebra $\mathcal{G} \subseteq \mathcal{F}$ (e.g. "only the first k of n observations," or "only a summary statistic").
- We then want the e-variable that is log-optimal *among those measurable w.r.t.* $\mathcal{G}$ – not the unrestricted optimum, which may not be computable from what we observe.

Proposition 3.22: conditional log-optimum = conditional likelihood ratio

If $\mathbb{Q} \ll \mathbb{P}$, the log-optimal e-variable w.r.t. $\mathcal{G}$ exists ($\mathbb{P}$-a.s. uniquely) and equals

$$E = \frac{\mathrm{d}\mathbb{Q}|_{\mathcal{G}}}{\mathrm{d}\mathbb{P}|_{\mathcal{G}}} = \mathbb{E}^{\mathbb{P}}\left[\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}} \,\middle|\, \mathcal{G}\right].$$

In words: restrict both distributions to what $\mathcal{G}$ can see, take their likelihood ratio – equivalently, average the full-information likelihood ratio over what $\mathcal{G}$ doesn't know.

3.6 Conditional log-optimality & log-optimal calibrators II

Corollary 3.23: the log-optimal calibrator

Suppose P is an exact p-variable ($\mathrm{Unif}[0,1]$ under $\mathbb{P}$) for testing $\mathbb{P}$ vs. $\mathbb{Q} \ll \mathbb{P}$, and $\mathbb{E}^{\mathbb{P}}[\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P} \mid P]$ is decreasing in P. Then the *log-optimal calibrator* f (recall: calibrators turn p-values into e-values) is

$$f(P) = \mathbb{E}^{\mathbb{P}}\left[\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}} \,\middle|\, P\right], \qquad f \equiv 0 \text{ on } (1, \infty).$$

Intuition: among all calibrators (Ch. 2), this is the one specifically tuned to be best against a particular known alternative $\mathbb{Q}$ – it is the conditional expectation of the likelihood ratio given only the p-value.

Section 3.7 – Intuition: what if $\mathcal{Q}$ isn't a single point?

- Theorem 3.20 needs a *simple* (single-distribution) alternative $\mathbb{Q}$ to write down $\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P}$ explicitly.
- Real problems usually have a *composite* $\mathcal{Q}$ (e.g. "some unknown $\theta_1 \neq \theta_0$"). We can't build the log-optimal e-variable directly – but we can get *asymptotically* just as good, without ever committing to a single guess.
- Two practical recipes: **mixtures** (average over the likely alternatives) and **plug-in** (estimate the alternative as data arrive, one step at a time).

3.7 Asymptotic log-optimality: mixtures and plug-in I

Goal: asymptotic log-optimality, Eq. (3.6)

For a composite $\mathcal{Q}$ (iid data, densities p, q per observation), we want e-variables $E_n = E(X^n)$ such that for every $\mathbb{Q} \in \mathcal{Q}$,

$$\lim_{n\to\infty} \frac{\mathbb{E}^{\mathbb{Q}^n}[\log E_n]}{n} \to \mathrm{KL}(\mathbb{Q} \,\|\, \mathbb{P}).$$

The right-hand side is the e-power of an all-knowing *oracle* that is handed the true $\mathbb{Q}$ in advance. Achieving this limit means our e-variable is (asymptotically) just as good as the oracle, without knowing $\mathbb{Q}$ up front.

3.7 Asymptotic log-optimality: mixtures and plug-in II

Method 1: mixtures, Eq. (3.7)

Pick a mixing ("prior") distribution ν over $\mathcal{Q}$, and define

$$E_n = \int \prod_{i=1}^{n} \frac{q(X_i)}{p(X_i)} \, \nu(\mathrm{d}q).$$

Read the order carefully: take the product of likelihood ratios *first*, then mix. This is a valid e-variable (average of e-variables). It works whenever ν has full support over $\mathcal{Q}$ and the mixture integral is tractable.

3.7 Asymptotic log-optimality: mixtures and plug-in III

Method 2: plug-in, Eq. (3.8)

Let $\mathbb{Q}_{i-1} \in \mathcal{Q}$ be an estimate of the alternative based on only the first $i-1$ observations (e.g. the Bayes posterior mean under prior ν), with density q_{i-1}. Define

$$E_n = \prod_{i=1}^{n} \frac{q_{i-1}(X_i)}{p(X_i)}.$$

Each factor is a "one-step-ahead" likelihood ratio using your current best guess; the product telescopes into a valid e-variable, and under mild conditions is asymptotically log-optimal whenever the corresponding mixture is.

Example 3.24: the t-test e-variable

$\mathcal{P} = \{N(0, \sigma^2) : \sigma > 0\}$, $\mathcal{Q} = \{N(\mu, \sigma^2) : \sigma > 0, \mu \neq 0\}$. With $S_n = \sum X_i$, $V_n = \sum X_i^2$, for any $c > 0$,

$$\sqrt{\frac{c^2}{n + c^2}} \left(\frac{(n + c^2)V_n}{(n + c^2)V_n - S_n^2} \right)^{n/2}$$

is a scale-invariant e-variable achieving (3.6). At $n = 1$ it equals 1 (zero e-power) – one observation alone can never provide t-test evidence. Remarkably, a *non-constant* e-variable with positive e-power at $n = 1$ does exist: $E_1 = |X_1| \exp((1 - (X_1 - a)^2)/2)$ for any fixed a, though it is not scale-invariant.

Example 3.25: changepoint e-value

n ordered observations; null: all iid $\mathbb{P}$. Alternative: for some unknown $k \in [n]$, the last k observations switch to a known $\mathbb{Q}$ while the rest stay $\mathbb{P}$. A natural *mixture over the unknown changepoint* k gives the e-value

$$\frac{1}{n}\sum_{k=1}^{n}\prod_{i=n-k+1}^{n}\frac{q(X_i)}{p(X_i)}.$$

This idea underlies the more general theory of *e-detectors* for sequential change detection (beyond this book's scope).

Python: mixture method for a composite alternative

```python
import numpy as np
from scipy import integrate

# H0: theta = 0.5 (fair coin). H1: theta in (0.5, 1), unknown -- composite!
# Mixture e-value: integrate the likelihood ratio over a prior on theta.
theta0 = 0.5

def mixture_evalue(heads, n, prior_lo=0.5, prior_hi=1.0):
    def integrand(theta):
        # likelihood ratio for the whole sample, at this theta
        return (theta**heads * (1-theta)**(n-heads)) / \
               (theta0**heads * (1-theta0)**(n-heads))
    val, _ = integrate.quad(integrand, prior_lo, prior_hi)
    return val / (prior_hi - prior_lo)   # uniform mixing density = 1/(hi-lo)

rng = np.random.default_rng(2)
n, theta_true = 100, 0.7
heads = rng.binomial(n, theta_true)
E = mixture_evalue(heads, n)
print(f"heads={heads}/{n}, mixture e-value={E:.3g}, log E={np.log(E):.3f}")
# compare to the (unavailable in practice) oracle LR e-value at theta=0.7
```

Even without knowing $\theta_1 = 0.7$ in advance, the mixture e-value grows large – illustrating asymptotic log-optimality without committing to a single alternative.

Section 3.8 – Intuition: why $\mathbb{E}^{\mathbb{Q}}[\log E]$, and not something else?

- So far we *defined* e-power as $\mathbb{E}^{\mathbb{Q}}[\log E]$ and showed it behaves well. But is it the *only* sensible choice, or just one of many?
- Section 3.8 shows: if you want a notion of "how powerful is E" that (a) scales sensibly when combined with prior evidence, and (b) agrees with the intuitive "asymptotic power 1 vs. 0" criterion of Proposition 3.12, then you are *forced* into using (a strictly increasing transform of) $\mathbb{E}^{\mathbb{Q}}[\log E]$.

3.8 Axiomatic justification of the e-power I

Setup

Let $\mathcal{X}$ be nonnegative random variables bounded above and away from 0 (technical convenience only). Fix an atomless $\mathbb{Q}$. We study mappings $\Lambda^{\mathbb{Q}} : \mathcal{X} \to \mathbb{R}$, candidates for "e-power," *without* requiring E to be a valid e-variable for any specific $\mathcal{P}$ – e-power is a property of behavior under $\mathbb{Q}$ alone.
$\Lambda^{\mathbb{Q}}(E_1) > \Lambda^{\mathbb{Q}}(E_2)$ should mean "E_1 is more powerful than E_2."

Axiom 3.26: Homogeneity

If $\Lambda^{\mathbb{Q}}(E_1) \geq \Lambda^{\mathbb{Q}}(E_2)$, then $\Lambda^{\mathbb{Q}}(e_0 E_1) \geq \Lambda^{\mathbb{Q}}(e_0 E_2)$ for any constant $e_0 > 0$.
Motivation: imagine a two-stage experiment where stage 1 already produced an e-value e_0. You must pick E_1 or E_2 for stage 2; the *effective* comparison is really between $e_0 E_1$ and $e_0 E_2$. Homogeneity says your stage-2 ranking should not be overturned just by relabeling/rescaling by the (fixed, already-observed) e_0.

Axiom 3.27: Asymptotic consistency

Let (E_n), (E'_n) be iid sequences under $\mathbb{Q}$. If $\Lambda^{\mathbb{Q}}(E_1) \geq \Lambda^{\mathbb{Q}}(E'_1)$, then for every $\alpha \in (0,1)$,

$$\mathbb{Q}\Big(\prod_{t=1}^{n} E'_t \geq \tfrac{1}{\alpha}\Big) \to 1 \implies \mathbb{Q}\Big(\prod_{t=1}^{n} E_t \geq \tfrac{1}{\alpha}\Big) \to 1.$$

Motivation: if the "more powerful" configuration by $\Lambda^{\mathbb{Q}}$ is pitted against one that already achieves asymptotic power 1 when repeated, the more powerful one must too. This encodes Proposition 3.12's growth-rate intuition as an axiom.

3.8 Axiomatic justification of the e-power III

Theorem 3.28: the representation theorem

$\Lambda^{\mathbb{Q}}$ satisfies Axioms 3.26–3.27 *if and only if*

$$\Lambda^{\mathbb{Q}}(E) = f\big(\mathbb{E}^{\mathbb{Q}}[\log E]\big), \qquad E \in \mathcal{X},$$

for some strictly increasing f.

Step-by-step proof sketch.

tep 1. Check $E \mapsto \mathbb{E}^{\mathbb{Q}}[\log E]$ itself satisfies both axioms (Axiom 3.27 via the strong law of large numbers); any strictly increasing transform of it therefore does too.

tep 2. (Converse, by contradiction) Suppose $\Lambda^{\mathbb{Q}}(E) > \Lambda^{\mathbb{Q}}(E')$ but $\mathbb{E}^{\mathbb{Q}}[\log E] < \mathbb{E}^{\mathbb{Q}}[\log E']$. Rescale by a well-chosen $c > 0$ (Axiom 3.26) so that $\log(cE) < 0 < \log(cE')$ in expectation.

tep 3. Apply the strong law of large numbers to iid copies: $\log \prod cE_t \to -\infty$ but $\log \prod cE'_t \to +\infty$ in $\mathbb{Q}$-probability – i.e. the "less powerful by $\Lambda^{\mathbb{Q}}$" sequence achieves asymptotic power 1 while the "more powerful" one achieves asymptotic power 0. This directly contradicts Axiom 3.27.

3.8 Axiomatic justification of the e-power IV

tep 4. Hence $\Lambda^{\mathbb{Q}}(E) > \Lambda^{\mathbb{Q}}(E') \Rightarrow \mathbb{E}^{\mathbb{Q}}[\log E] \geq \mathbb{E}^{\mathbb{Q}}[\log E']$; a further tie-breaking argument (using Axiom 3.26 again, and monotonicity/bounded-variation of $c \mapsto \Lambda^{\mathbb{Q}}(cE)$) upgrades this to strict inequality, showing $\Lambda^{\mathbb{Q}}$ and $E \mapsto \mathbb{E}^{\mathbb{Q}}[\log E]$ induce the *same ordering* – hence the representation.

Punch line

If you accept Axioms 3.26–3.27 as reasonable requirements for any sensible notion of "e-power," **then** $\mathbb{E}^{\mathbb{Q}}[\log E]$ (up to relabeling by a strictly increasing f) is the *only* possible choice. A natural such transform is the *geometric mean* $\exp(\mathbb{E}^{\mathbb{Q}}[\log E])$: it is nonnegative, and exceeds 1 exactly when e-power is positive.

Proposition 3.29: sanity-check properties of $\Lambda^{\mathbb{Q}}$

For an e-variable E: (i) $\Lambda^{\mathbb{Q}}(E) > 0 \Rightarrow \Lambda^{\mathbb{Q}}(1-\lambda+\lambda E) > 0$ for all $\lambda \in (0,1]$; (ii) same with "≥ 0"; (iii) $\Lambda^{\mathbb{Q}}(1-\lambda+\lambda E)$ is always well-defined for $\lambda \in [0,1]$; (iv) if $\Lambda^{\mathbb{Q}}(E) < \infty$, then $\Lambda^{\mathbb{Q}}(1-\lambda+\lambda E) \in \mathbb{R}$ for all $\lambda \in [0,1]$. (Compare Theorem 3.14 – shrinking E towards 1 preserves the sign of e-power.)

Chapter 3 summary

- **Powered** tests/e-variables/p-variables (3.1) are unified by Theorem 3.9: any one form of testability gives you all the others (under mild conditions).
- **E-power** $\mathbb{E}^{\mathbb{Q}}[\log E]$ (3.3) is the right measure of "how good" an e-variable is for repeated/sequential use – it is the asymptotic growth rate of cumulative evidence (Kelly criterion).
- Existence of powered procedures is checkable via a **total-variation separation** condition (3.4, Theorem 3.17).
- **Log-optimality** (3.5): for simple $\mathbb{P}$ vs. simple $\mathbb{Q}$, the best possible e-variable is exactly the **likelihood ratio**, and its e-power is exactly the **KL divergence** $\mathrm{KL}(\mathbb{Q} \,\|\, \mathbb{P})$ (Theorem 3.20).
- When restricted to observable data (3.6), the log-optimal e-variable is a **conditional** likelihood ratio; this gives the log-optimal calibrator for p-values.
- For composite alternatives (3.7), **mixture** and **plug-in** constructions achieve the oracle's KL growth rate asymptotically, without knowing $\mathbb{Q}$ in advance.
- The choice of $\mathbb{E}^{\mathbb{Q}}[\log E]$ as "e-power" is not arbitrary: it is the *unique* (up to relabeling) notion satisfying two natural axioms (3.8, Theorem 3.28).
- **Running numeral:** testing a coin ($\theta_0 = 0.5$ vs. $\theta_1 = 0.7$), the log-optimal e-variable has e-power $\mathrm{KL}(0.7\|0.5) = 0.0823$ nats/flip, versus 0.0607 for a naive mis-targeted e-variable – an $\approx 8.7\times$ faster-growing e-process after 100 flips.